

AMERICAN TERAWATT

HVDC Configurations

System Topologies

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AC Loadflow

AC loadflow calculations

Symbols

Sym.	Description
P	Active power
V_{LL}	Line-to-line (phase-phase) voltage
V_{LN}	Line-to-neutral (phase-ground) voltage

Abbreviations

Abbr.	Meaning
AC	Alternating current
DC	Direct current
HVDC	High-voltage direct current
OHL	Overhead line

Current calculation

- The current in the on the DC line or busbar is calculated by:

$$I = \frac{P}{\sqrt{3} \cdot V_{LL}}$$

- Note that for current calculation the phase-phase voltage V_{LL} is used
- The maximum current is limited not only by the DC line capability but also by equipment on the DC line such as circuit breakers. The maximum current rating for circuit breakers is 5000A (Hitachi Energy, GE).